

SOLIDWORKS Document Manager API (.NET) — Reference (1/1/19) ISwDMDimXpertParallelismGeoTol- ISwDMDimXpertSurfaceProfileGeoTol

SOLIDWORKS API Documentation (converted from CHM)

ISwDMDimXpertParallelismGeoTol (Type)

ISwDMDimXpertParallelismGeoTol Interface

[See Also](#) [Members](#) [Example](#)



[SolidWorks.Interop.swdocumentmgr Namespace](#) : ISwDMDimXpertParallelismGeoTol Interface

Allows you to access DimXpert parallelism geometric tolerance information.

NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

.NET Syntax

Visual Basic (Declaration)

```
Public Interface ISwDMDimXpertParallelismGeoTol
```

Visual Basic (Usage)  [Copy Code](#)

```
Dim instance As ISwDMDimXpertParallelismGeoTol
```

C#

```
public interface ISwDMDimXpertParallelismGeoTol
```

C++/CLI

```
public interface class ISwDMDimXpertParallelismGeoTol
```

Visual Basic for Applications (VBA) Syntax

See [SwDMDimXpertParallelismGeoTol](#).

Example

[Get DimXpert Tolerance8 \(VB.NET\)](#)
[Get DimXpert Tolerance8 \(C#\)](#)

Remarks

This interface is one of three orientation interfaces described by [ISwDMDimXpertOrientationGeoTol](#).
You can access more general information from [ISwDMDimXpertGeometricTolerance](#).

Accessors

[ISwDMDimXpertPart::GetAnnotations](#) and [ISwDMDimXpertPart::IGetAnnotations](#)

See Also

[ISwDMDimXpertParallelismGeoTol Members](#)
[SolidWorks.Interop.swdocumentmgr Namespace](#)

ISwDMDimXpertParallelismGeoTol Interface Members

[See Also](#)



[SolidWorks.Interop.swdocumentmgr Namespace](#) : ISwDMDimXpertParallelismGeoTol Interface

The following tables list the members exposed by [ISwDMDimXpertParallelismGeoTol](#).

See Also

[ISwDMDimXpertParallelismGeoTol Interface](#)
[SolidWorks.Interop.swdocumentmgr Namespace](#)

ISwDMDimXpertPart (Type)

ISwDMDimXpertPart Interface

[See Also](#) [Members](#) [Example](#)



[SolidWorks.Interop.swdocumentmgr.Namespace](#) : ISwDMDimXpertPart Interface

Allows you to access the geometric and dimension tolerance information that underpins the DimXpert functionality, features, and annotations of a given model.

NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

.NET Syntax

Visual Basic (Declaration)

Public Interface ISwDMDimXpertPart

Visual Basic (Usage)

Dim instance As [ISwDMDimXpertPart](#)

C#

public interface ISwDMDimXpertPart

C++/CLI

public interface class ISwDMDimXpertPart

Visual Basic for Applications (VBA) Syntax

See [SwDMDimXpertPart](#).

Example

[Get DimXpert Block Tolerance \(VB.NET\)](#)
[Get DimXpert Block Tolerance \(C#\)](#)

Accessors

[ISwDMConfiguration12::DimXpertPart property](#)

See Also

[ISwDMDimXpertPart Members](#)
[SolidWorks.Interop.swdocumentmgr.Namespace](#)

ISwDMDimXpertPart Interface Methods

[See Also](#)



[SolidWorks.Interop.swdocumentmgr.Namespace](#) : ISwDMDimXpertPart Interface

For a list of all members of this type, see [ISwDMDimXpertPart members](#).

Public Methods

Name	Description
 GetAnnotationCount	Gets the total number of DimXpert annotations in the model.
 GetAnnotations	Gets all of the DimXpert annotations in the model.
 GetBlockTolerances	Gets all of the DimXpert block tolerances in the model.
 GetFeatureCount	Gets the total number of DimXpert features in the model.

 GetFeatures	Gets all of the DimXpert features in the model.
 IGetAnnotations	Gets all of the DimXpert annotations in the model.
 IGetFeatures	Gets all of the DimXpert features in the model.

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 See Also

[ISwMDimXpertPart Interface](#)
[SolidWorks.Interop.swdocumentmgr Namespace](#)

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ISwMDimXpertPart Interface Members

[See Also](#) [Methods](#)



[SolidWorks.Interop.swdocumentmgr Namespace](#) : ISwMDimXpertPart Interface

The following tables list the members exposed by [ISwMDimXpertPart](#).

 Public Methods

Name	Description
 GetAnnotationCount	Gets the total number of DimXpert annotations in the model.
 GetAnnotations	Gets all of the DimXpert annotations in the model.
 GetBlockTolerances	Gets all of the DimXpert block tolerances in the model.
 GetFeatureCount	Gets the total number of DimXpert features in the model.
 GetFeatures	Gets all of the DimXpert features in the model.
 IGetAnnotations	Gets all of the DimXpert annotations in the model.
 IGetFeatures	Gets all of the DimXpert features in the model.

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 See Also

[ISwMDimXpertPart Interface](#)
[SolidWorks.Interop.swdocumentmgr Namespace](#)

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GetAnnotationCount Method (ISwMDimXpertPart)

[See Also](#)



[SolidWorks.Interop.swdocumentmgr Namespace](#) > [ISwMDimXpertPart Interface](#) : GetAnnotationCount Method (ISwMDimXpertPart)

Gets the total number of DimXpert annotations in the model.

 .NET Syntax

Visual Basic (Declaration)

```
Function GetAnnotationCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ISwMDimXpertPart  
Dim value As System.Integer  
  
value = instance.GetAnnotationCount()
```

C#

```
System.int GetAnnotationCount()
```

C++/CLI

```
System.int GetAnnotationCount();
```

Return Value

Number of annotations

Visual Basic for Applications (VBA) Syntax

See [SwMDimXpertPart::GetAnnotationCount](#).

Example

See the examples on the interface page.

See Also

[ISwMDimXpertPart Interface](#)
[ISwMDimXpertPart Members](#)

Availability

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GetAnnotations Method (ISwMDimXpertPart)

[See Also](#)



[SolidWorks.Interop.swdocumentmgr Namespace](#) > [ISwMDimXpertPart Interface](#) : GetAnnotations Method (ISwMDimXpertPart)

Gets all of the DimXpert annotations in the model.

.NET Syntax

Visual Basic (Declaration)

```
Function GetAnnotations() As System.Object
```

Visual Basic (Usage)



```
Dim instance As ISwMDimXpertPart  
Dim value As System.Object
```

```
value = instance.GetAnnotations()
```

C#

```
System.Object GetAnnotations()
```

C++/CLI

```
System.Object^ GetAnnotations();
```

Return Value

Array of [ISwMDimXpertAnnotations](#)

Visual Basic for Applications (VBA) Syntax

See [SwMDimXpertPart::GetAnnotations](#).

Example

See the examples on the interface page.

See Also

[ISwMDimXpertPart Interface](#)
[ISwMDimXpertPart Members](#)
[ISwMDimXpertPart::IGetAnnotations Method](#)

Availability

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GetBlockTolerances Method (ISwMDimXpertPart)

[See Also](#)



[SolidWorks.Interop.swdocumentmgr Namespace](#) > [ISwMDimXpertPart Interface](#) : GetBlockTolerances Method (ISwMDimXpertPart)

Gets all of the DimXpert block tolerances in the model.

.NET Syntax

Visual Basic (Declaration)

```
Function GetBlockTolerances() As SwMDimXpertBlockTolerances
```

Visual Basic (Usage)



```
Dim instance As ISwMDimXpertPart
Dim value As SwMDimXpertBlockTolerances

value = instance.GetBlockTolerances()
```

C#

```
SwMDimXpertBlockTolerances GetBlockTolerances()
```

C++/CLI

```
SwMDimXpertBlockTolerances^ GetBlockTolerances();
```

Return Value

[ISwMDimXpertBlockTolerances](#)

Visual Basic for Applications (VBA) Syntax

See [SwMDimXpertPart::GetBlockTolerances](#).

Example

See the examples on the interface page.

See Also

[ISwMDimXpertPart Interface](#)
[ISwMDimXpertPart Members](#)

Availability

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GetFeatureCount Method (ISwMDimXpertPart)

[See Also](#)



[SolidWorks.Interop.swdocumentmgr Namespace](#) > [ISwMDimXpertPart Interface](#) : GetFeatureCount Method (ISwMDimXpertPart)

Gets the total number of DimXpert features in the model.

.NET Syntax

Visual Basic (Declaration)

```
Function GetFeatureCount() As System.Integer
```

```
Dim instance As ISwDMDimXpertPart
Dim value As System.Integer

value = instance.GetFeatureCount()
```

C#

```
System.int GetFeatureCount()
```

C++/CLI

```
System.int GetFeatureCount();
```

Return Value

Number of features

Visual Basic for Applications (VBA) SyntaxSee [SwDMDimXpertPart::GetFeatureCount](#). **Example**

See the examples on the interface page.

See Also

[ISwDMDimXpertPart Interface](#)
[ISwDMDimXpertPart Members](#)

Availability

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[Send comments](#) on this topic.**GetFeatures Method (ISwDMDimXpertPart)**[See Also](#)[SolidWorks.Interop.swdocumentmgr Namespace](#) > [ISwDMDimXpertPart Interface](#) : GetFeatures Method (ISwDMDimXpertPart)

Gets all of the DimXpert features in the model.

.NET Syntax

Visual Basic (Declaration)

```
Function GetFeatures() As System.Object
```

Visual Basic (Usage)

```
Dim instance As ISwDMDimXpertPart
Dim value As System.Object

value = instance.GetFeatures()
```

C#

```
System.object GetFeatures()
```

C++/CLI

```
System.Object^ GetFeatures();
```

Return ValueArray of [ISwDMDimXpertFeatures](#)

Visual Basic for Applications (VBA) Syntax

See [SwDMDimXpertPart::GetFeatures](#).

Example

See the examples on the interface page.

See Also

[ISwDMDimXpertPart Interface](#)
[ISwDMDimXpertPart Members](#)
[ISwDMDimXpertPart::IGetFeatures Method](#)

Availability

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IGetAnnotations Method (ISwDMDimXpertPart)

[See Also](#)



[SolidWorks.Interop.swdocumentmgr Namespace](#) > [ISwDMDimXpertPart Interface](#) : IGetAnnotations Method (ISwDMDimXpertPart)

Gets all of the DimXpert annotations in the model.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetAnnotations(  
    ByVal Count As System.Integer _  
) As SwDMDimXpertAnnotation
```

Visual Basic (Usage)



```
Dim instance As ISwDMDimXpertPart  
Dim Count As System.Integer  
Dim value As SwDMDimXpertAnnotation  
  
value = instance.IGetAnnotations(Count)
```

C#

```
SwDMDimXpertAnnotation IGetAnnotations(  
    System.int Count  
)
```

C++/CLI

```
SwDMDimXpertAnnotation^ IGetAnnotations(  
&    System.int Count  
)
```

Parameters

Count
Number of annotations

Return Value

- in-process, unmanaged C++: Pointer to an array of [ISwDMDimXpertAnnotation](#)
- VBA, VB.NET, C#, and C++/CLI: Not supported

Remarks

Use [ISwDMDimXpertPart::GetAnnotationCount](#) to populate the Count argument.

See Also

[ISwDMDimXpertPart Interface](#)
[ISwDMDimXpertPart Members](#)
[ISwDMDimXpertPart::GetAnnotations Method](#)

Availability

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IGetFeatures Method (ISwDMDimXpertPart)

[See Also](#)



[SolidWorks.Interop.swdocumentmgr Namespace](#) > [ISwDMDimXpertPart Interface](#) : IGetFeatures Method (ISwDMDimXpertPart)

Gets all of the DimXpert features in the model.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetFeatures( _  
    ByVal Count As System.Integer _  
) As SwDMDimXpertFeature
```

Visual Basic (Usage)



```
Dim instance As ISwDMDimXpertPart  
Dim Count As System.Integer  
Dim value As SwDMDimXpertFeature  
  
value = instance.IGetFeatures(Count)
```

C#

```
SwDMDimXpertFeature IGetFeatures(  
    System.int Count  
)
```

C++/CLI

```
SwDMDimXpertFeature^ IGetFeatures(  
& System.int Count  
)
```

Parameters

Count
Number of features

Return Value

- in-process, unmanaged C++: Pointer to an array of [ISwDMDimXpertFeature](#)
- VBA, VB.NET, C#, and C++/CLI: Not supported

Remarks

Use [ISwDMDimXpertPart::GetFeatureCount](#) to populate the Count argument.

See Also

[ISwDMDimXpertPart Interface](#)
[ISwDMDimXpertPart Members](#)
[ISwDMDimXpertPart::GetFeatures Method](#)

Availability

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ISwDMDimXpertPatternAngleBetweenDimTol (Type)

ISwDMDimXpertPatternAngleBetweenDimTol Interface

[See Also](#) [Members](#)



[SolidWorks.Interop.swdocumentmgr.Namespace](#) : ISwDMDimXpertPatternAngleBetweenDimTol Interface

Allows you to access DimXpert a pattern angle dimension tolerances.

NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

.NET Syntax

Visual Basic (Declaration)

```
Public Interface ISwDMDimXpertPatternAngleBetweenDimTol
```

Visual Basic (Usage)



```
Dim instance As ISwDMDimXpertPatternAngleBetweenDimTol
```

C#

```
public interface ISwDMDimXpertPatternAngleBetweenDimTol
```

C++/CLI

```
public interface class ISwDMDimXpertPatternAngleBetweenDimTol
```

Visual Basic for Applications (VBA) Syntax

See [SwDMDimXpertPatternAngleBetweenDimTol](#).

Remarks

You can access more general information from [ISwDMDimXpertDimensionTolerance](#).

Accessors

[ISwDMDimXpertPart::GetAnnotations](#) and [ISwDMDimXpertPart::!GetAnnotations](#)

See Also

[ISwDMDimXpertPatternAngleBetweenDimTol Members](#)
[SolidWorks.Interop.swdocumentmgr.Namespace](#)

ISwDMDimXpertPatternAngleBetweenDimTol Interface Members

[See Also](#)



[SolidWorks.Interop.swdocumentmgr.Namespace](#) : ISwDMDimXpertPatternAngleBetweenDimTol Interface

The following tables list the members exposed by [ISwDMDimXpertPatternAngleBetweenDimTol](#).

See Also

[ISwDMDimXpertPatternAngleBetweenDimTol Interface](#)
[SolidWorks.Interop.swdocumentmgr.Namespace](#)

ISwDMDimXpertPatternFeature (Type)

ISwDMDimXpertPatternFeature Interface

[See Also](#) [Members](#) [Example](#)



[SolidWorks.Interop.swdocumentmgr Namespace](#) : ISwDMDimXpertPatternFeature Interface

Allows you to access specific information about DimXpert patterns.

NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

.NET Syntax

Visual Basic (Declaration)

Public Interface ISwDMDimXpertPatternFeature

Visual Basic (Usage)

Dim instance As [ISwDMDimXpertPatternFeature](#)

C#

public interface ISwDMDimXpertPatternFeature

C++/CLI

public interface class ISwDMDimXpertPatternFeature

Visual Basic for Applications (VBA) Syntax

See [SwDMDimXpertPatternFeature](#).

Example

[Get DimXpert Feature \(VB.NET\)](#)
[Get DimXpert Feature \(C#\)](#)

Remarks

You can access more general information from [ISwDMDimXpertFeature](#).

Accessors

[ISwDMDimXpertPart::GetFeatures](#) and [ISwDMDimXpertPart::IGetFeatures](#)

See Also

[ISwDMDimXpertPatternFeature Members](#)
[SolidWorks.Interop.swdocumentmgr Namespace](#)

ISwDMDimXpertPatternFeature Interface Members

[See Also](#) [Methods](#)



[SolidWorks.Interop.swdocumentmgr Namespace](#) : ISwDMDimXpertPatternFeature Interface

The following tables list the members exposed by [ISwDMDimXpertPatternFeature](#).

Public Methods

Name	Description
 GetSubFeatureCount	Gets the number of sub-features in this DimXpert pattern feature.
 GetSubFeatures	Gets all of the sub-features of this DimXpert pattern feature.
IGetSubFeatures	Gets all of the sub-features in this DimXpert pattern feature.



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See Also

[ISwDMDimXpertPatternFeature Interface](#)
[SolidWorks.Interop.swdocumentmgr Namespace](#)

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ISwDMDimXpertPatternFeature Interface Methods

[See Also](#)



[SolidWorks.Interop.swdocumentmgr Namespace](#) : ISwDMDimXpertPatternFeature Interface

For a list of all members of this type, see [ISwDMDimXpertPatternFeature members](#).

Public Methods

Name	Description
 GetSubFeatureCount	Gets the number of sub-features in this DimXpert pattern feature.
 GetSubFeatures	Gets all of the sub-features of this DimXpert pattern feature.
 IGetSubFeatures	Gets all of the sub-features in this DimXpert pattern feature.

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See Also

[ISwDMDimXpertPatternFeature Interface](#)
[SolidWorks.Interop.swdocumentmgr Namespace](#)

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GetSubFeatureCount Method (ISwDMDimXpertPatternFeature)

[See Also](#)



[SolidWorks.Interop.swdocumentmgr Namespace](#) > [ISwDMDimXpertPatternFeature Interface](#) : GetSubFeatureCount Method (ISwDMDimXpertPatternFeature)

Gets the number of sub-features in this DimXpert pattern feature.

.NET Syntax

Visual Basic (Declaration)

```
Function GetSubFeatureCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ISwDMDimXpertPatternFeature  
Dim value As System.Integer  
  
value = instance.GetSubFeatureCount()
```

C#

```
System.int GetSubFeatureCount()
```

C++/CLI

```
System.int GetSubFeatureCount();
```

Return Value

Number of sub-features in this pattern feature

Visual Basic for Applications (VBA) Syntax

See [SwDMDimXpertPatternFeature::GetSubFeatureCount](#).

Example

See the examples on the interface page.

See Also

[ISwDMDimXpertPatternFeature Interface](#)
[ISwDMDimXpertPatternFeature Members](#)

Availability

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GetSubFeatures Method (ISwDMDimXpertPatternFeature)

[See Also](#)



[SolidWorks.Interop.swdocumentmgr Namespace](#) > [ISwDMDimXpertPatternFeature Interface](#) : GetSubFeatures Method (ISwDMDimXpertPatternFeature)

Gets all of the sub-features of this DimXpert pattern feature.

.NET Syntax

Visual Basic (Declaration)

```
Function GetSubFeatures() As System.Object
```

Visual Basic (Usage)



```
Dim instance As ISwDMDimXpertPatternFeature  
Dim value As System.Object  
  
value = instance.GetSubFeatures()
```

C#

```
System.Object GetSubFeatures()
```

C++/CLI

```
System.Object^ GetSubFeatures();
```

Return Value

Array of [ISwDMDimXpertFeatures](#)

Visual Basic for Applications (VBA) Syntax

See [SwDMDimXpertPatternFeature::GetSubFeatures](#).

Example

See the examples on the interface page.

See Also

[ISwDMDimXpertPatternFeature Interface](#)
[ISwDMDimXpertPatternFeature Members](#)
[ISwDMDimXpertPatternFeature::IGetSubFeatures Method](#)

Availability

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IGetSubFeatures Method (ISwDMDimXpertPatternFeature)

[See Also](#)



[SolidWorks.Interop.swdocumentmgr Namespace](#) > [ISwMDimXpertPatternFeature Interface](#) : IGetSubFeatures Method (ISwMDimXpertPatternFeature)

Gets all of the sub-features in this DimXpert pattern feature.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetSubFeatures( _  
    ByVal Count As System.Integer _  
) As SwMDimXpertFeature
```

Visual Basic (Usage)



```
Dim instance As ISwMDimXpertPatternFeature  
Dim Count As System.Integer  
Dim value As SwMDimXpertFeature  
  
value = instance.IGetSubFeatures(Count)
```

C#

```
SwMDimXpertFeature IGetSubFeatures(  
    System.int Count  
)
```

C++/CLI

```
SwMDimXpertFeature^ IGetSubFeatures(  
&    System.int Count  
)
```

Parameters

Count

Number of sub-features in this pattern

Return Value

- in-process, unmanaged C++: Pointer to an array of [ISwMDimXpertFeature](#)
- VBA, VB.NET, C#, and C++/CLI: Not supported

Remarks

Use [ISwMDimXpertPatternFeature::GetSubFeatureCount](#) to populate the Count argument.

See Also

[ISwMDimXpertPatternFeature Interface](#)
[ISwMDimXpertPatternFeature Members](#)
[ISwMDimXpertPatternFeature::GetSubFeatures Method](#)

Availability

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ISwDMDimXpertPerpendicularityGeoTol (Type)

ISwDMDimXpertPerpendicularityGeoTol Interface

[See Also](#) [Members](#) [Example](#)



[SolidWorks.Interop.swdocumentmgr Namespace](#) : ISwDMDimXpertPerpendicularityGeoTol Interface

Allows you to access DimXpert perpendicularity geometric tolerance information.

NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

.NET Syntax

Visual Basic (Declaration)

```
Public Interface ISwDMDimXpertPerpendicularityGeoTol
```

Visual Basic (Usage)



```
Dim instance As ISwDMDimXpertPerpendicularityGeoTol
```

C#

```
public interface ISwDMDimXpertPerpendicularityGeoTol
```

C++/CLI

```
public interface class ISwDMDimXpertPerpendicularityGeoTol
```

Visual Basic for Applications (VBA) Syntax

See [SwDMDimXpertPerpendicularityGeoTol](#).

Example

[Get DimXpert Tolerance3 \(VB.NET\)](#)
[Get DimXpert Tolerance3 \(C#\)](#)

Remarks

This interface is one of three orientation interfaces described by [ISwDMDimXpertOrientationGeoTol](#).
You can access more general information from [ISwDMDimXpertGeometricTolerance](#).

Accessors

[ISwDMDimXpertPart::GetAnnotations](#) and [ISwDMDimXpertPart::IGetAnnotations](#)

See Also

[ISwDMDimXpertPerpendicularityGeoTol Members](#)
[SolidWorks.Interop.swdocumentmgr Namespace](#)

ISwDMDimXpertPerpendicularityGeoTol Interface Members

[See Also](#)



[SolidWorks.Interop.swdocumentmgr Namespace](#) : ISwDMDimXpertPerpendicularityGeoTol Interface

The following tables list the members exposed by [ISwDMDimXpertPerpendicularityGeoTol](#).

See Also

[ISwDMDimXpertPerpendicularityGeoTol Interface](#)
[SolidWorks.Interop.swdocumentmgr Namespace](#)

ISwDMDimXpertPlaneFeature (Type)

ISwDMDimXpertPlaneFeature Interface

[See Also](#) [Members](#) [Example](#)



[SolidWorks.Interop.swdocumentmgr Namespace](#) : ISwDMDimXpertPlaneFeature Interface

Allows you to access specific information about DimXpert planes.

NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

.NET Syntax

Visual Basic (Declaration)

```
Public Interface ISwDMDimXpertPlaneFeature
```

Visual Basic (Usage)



```
Dim instance As ISwDMDimXpertPlaneFeature
```

C#

```
public interface ISwDMDimXpertPlaneFeature
```

C++/CLI

```
public interface class ISwDMDimXpertPlaneFeature
```

Visual Basic for Applications (VBA) Syntax

See [SwDMDimXpertPlaneFeature](#).

Example

[Get DimXpert Feature \(VB.NET\)](#)
[Get DimXpert Feature \(C#\)](#)

Remarks

You can access more general information from [ISwDMDimXpertFeature](#).

Accessors

[ISwDMDimXpertCompoundClosedSlot3dFeature::GetPlaneFeatures](#)
[ISwDMDimXpertCompoundNotchFeature::GetPlaneFeatures](#)
[ISwDMDimXpertCompoundWidthFeature::GetPlaneFeatures](#)
[ISwDMDimXpertPart::GetFeatures](#) and [ISwDMDimXpertPart::IGetFeatures](#)

See Also

[ISwDMDimXpertPlaneFeature Members](#)
[SolidWorks.Interop.swdocumentmgr Namespace](#)

ISwDMDimXpertPlaneFeature Interface Methods

[See Also](#)



[SolidWorks.Interop.swdocumentmgr Namespace](#) : ISwDMDimXpertPlaneFeature Interface

For a list of all members of this type, see [ISwDMDimXpertPlaneFeature members](#).

Public Methods

Name	Description
------	-------------

	GetNominalPlane	Gets the coordinates and vector for this DimXpert plane.
--	---------------------------------	--

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 See Also

[ISwMDimXpertPlaneFeature Interface](#)
[SolidWorks.Interop.swdocumentmgr Namespace](#)

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ISwMDimXpertPlaneFeature Interface Members

[See Also](#) [Methods](#)

	SolidWorks.Interop.swdocumentmgr Namespace : ISwMDimXpertPlaneFeature Interface
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The following tables list the members exposed by [ISwMDimXpertPlaneFeature](#).

 Public Methods

Name	Description
 GetNominalPlane	Gets the coordinates and vector for this DimXpert plane.

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 See Also

[ISwMDimXpertPlaneFeature Interface](#)
[SolidWorks.Interop.swdocumentmgr Namespace](#)

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GetNominalPlane Method (ISwMDimXpertPlaneFeature)

[See Also](#)

		SolidWorks.Interop.swdocumentmgr Namespace > ISwMDimXpertPlaneFeature Interface : GetNominalPlane Method (ISwMDimXpertPlaneFeature)
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Gets the coordinates and vector for this DimXpert plane.

 .NET Syntax

Visual Basic (Declaration)

```
Function GetNominalPlane( _
    ByRef X As System.Double, _
    ByRef Y As System.Double, _
    ByRef Z As System.Double, _
    ByRef I As System.Double, _
    ByRef J As System.Double, _
    ByRef K As System.Double _
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISwMDimXpertPlaneFeature
Dim X As System.Double
Dim Y As System.Double
Dim Z As System.Double
Dim I As System.Double
Dim J As System.Double
Dim K As System.Double
Dim value As System.Boolean
```

```
value = instance.GetNominalPlane(X, Y, Z, I, J, K)
```

C#

```
System.bool GetNominalPlane(
    out System.double X,
    out System.double Y,
    out System.double Z,
    out System.double I,
```

```
    out System.double J,  
    out System.double K  
)
```

C++/CLI

```
System.bool GetNominalPlane(  
& [Out] System.double X,  
& [Out] System.double Y,  
& [Out] System.double Z,  
& [Out] System.double I,  
& [Out] System.double J,  
& [Out] System.double K  
)
```

Parameters

X
x coordinate of the plane
Y
y coordinate of the plane
Z
z coordinate of the plane
I
i component of the direction vector of the plane
J
j component of the direction vector of the plane
K
k component of the direction vector of the plane

Return Value

True if method call is successful; false otherwise

Visual Basic for Applications (VBA) Syntax

See [SwMDimXpertPlaneFeature::GetNominalPlane](#).

Example

See the examples on the interface page.

See Also

[ISwMDimXpertPlaneFeature Interface](#)
[ISwMDimXpertPlaneFeature Members](#)

Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

ISwDMDimXpertPositionGeoTol (Type)

ISwDMDimXpertPositionGeoTol Interface

[See Also](#) [Members](#) [Example](#)



[SolidWorks.Interop.swdocumentmgr Namespace](#) : ISwDMDimXpertPositionGeoTol Interface

Allows you to access DimXpert position geometric tolerance information.

NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

.NET Syntax

Visual Basic (Declaration)

```
Public Interface ISwDMDimXpertPositionGeoTol
```

Visual Basic (Usage)

```
Dim instance As ISwDMDimXpertPositionGeoTol
```

C#

```
public interface ISwDMDimXpertPositionGeoTol
```

C++/CLI

```
public interface class ISwDMDimXpertPositionGeoTol
```

Visual Basic for Applications (VBA) Syntax

See [SwDMDimXpertPositionGeoTol](#).

Example

[Get DimXpert Tolerance3 \(VB.NET\)](#)
[Get DimXpert Tolerance3 \(C#\)](#)

Remarks

You can access more general information from [ISwDMDimXpertGeometricTolerance](#).

Accessors

[ISwDMDimXpertPart::GetAnnotations](#) and [ISwDMDimXpertPart::!GetAnnotations](#)

See Also

[ISwDMDimXpertPositionGeoTol Members](#)
[SolidWorks.Interop.swdocumentmgr Namespace](#)

ISwDMDimXpertPositionGeoTol Interface Methods

[See Also](#)



[SolidWorks.Interop.swdocumentmgr Namespace](#) : ISwDMDimXpertPositionGeoTol Interface

For a list of all members of this type, see [ISwDMDimXpertPositionGeoTol members](#).

Public Methods

Name	Description
GetMaxTolerance	Gets the maximum tolerance for this DimXpert position geometric tolerance.
GetPlanarZoneVector	Gets the planar zone vector that is used to compute this DimXpert position geometric tolerance.
GetProjectedZone	Gets the projected zone value of this DimXpert position geometric tolerance.



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See Also

[ISwDMDimXpertPositionGeoTol Interface](#)
[SolidWorks.Interop.swdocumentmgr Namespace](#)

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ISwDMDimXpertPositionGeoTol Interface Members

[See Also](#) [Properties](#) [Methods](#)



[SolidWorks.Interop.swdocumentmgr Namespace](#) : ISwDMDimXpertPositionGeoTol Interface

The following tables list the members exposed by [ISwDMDimXpertPositionGeoTol](#).

Public Properties

Name	Description
 Modifier	Gets the material condition modifier for this DimXpert position geometric tolerance.
 ZoneType	Gets the zone type of this DimXpert position geometric tolerance.

[Top](#)

Public Methods

Name	Description
 GetMaxTolerance	Gets the maximum tolerance for this DimXpert position geometric tolerance.
 GetPlanarZoneVector	Gets the planar zone vector that is used to compute this DimXpert position geometric tolerance.
 GetProjectedZone	Gets the projected zone value of this DimXpert position geometric tolerance.

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See Also

[ISwDMDimXpertPositionGeoTol Interface](#)
[SolidWorks.Interop.swdocumentmgr Namespace](#)

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ISwDMDimXpertPositionGeoTol Interface Properties

[See Also](#)



[SolidWorks.Interop.swdocumentmgr Namespace](#) : ISwDMDimXpertPositionGeoTol Interface

For a list of all members of this type, see [ISwDMDimXpertPositionGeoTol members](#).

Public Properties

Name	Description
 Modifier	Gets the material condition modifier for this DimXpert position geometric tolerance.
 ZoneType	Gets the zone type of this DimXpert position geometric tolerance.

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See Also

[ISwDMDimXpertPositionGeoTol Interface](#)
[SolidWorks.Interop.swdocumentmgr Namespace](#)

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GetMaxTolerance Method (ISwDMDimXpertPositionGeoTol)

[See Also](#)



[SolidWorks.Interop.swdocumentmgr Namespace](#) > [ISwDMDimXpertPositionGeoTol Interface](#) : GetMaxTolerance Method (ISwDMDimXpertPositionGeoTol)

Gets the maximum tolerance for this DimXpert position geometric tolerance.

.NET Syntax

Visual Basic (Declaration)

```
Function GetMaxTolerance( _  
    ByRef IsMax As System.Boolean, _  
    ByRef Tolerance As System.Double _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISwDMDimXpertPositionGeoTol  
Dim IsMax As System.Boolean  
Dim Tolerance As System.Double  
Dim value As System.Boolean  
  
value = instance.GetMaxTolerance(IsMax, Tolerance)
```

C#

```
System.bool GetMaxTolerance(  
    out System.bool IsMax,  
    out System.double Tolerance  
)
```

C++/CLI

```
System.bool GetMaxTolerance(  
& [Out] System.bool IsMax,  
& [Out] System.double Tolerance  
)
```

Parameters

IsMax
True, if maximum tolerance is set; false, otherwise
Tolerance
Maximum tolerance

Return Value

True if method call is successful; false otherwise

Visual Basic for Applications (VBA) Syntax

See [SwDMDimXpertPositionGeoTol::GetMaxTolerance](#).

Example

See the examples on the interface page.

See Also

[ISwDMDimXpertPositionGeoTol Interface](#)
[ISwDMDimXpertPositionGeoTol Members](#)

Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SOLIDWORKS Document Manager API Help

[Send comments on this topic.](#)

GetPlanarZoneVector Method (ISwDMDimXpertPositionGeoTol)

[See Also](#)



[SolidWorks.Interop.swdocumentmgr Namespace](#) > [ISwDMDimXpertPositionGeoTol Interface](#) : GetPlanarZoneVector Method (ISwDMDimXpertPositionGeoTol)

Gets the planar zone vector that is used to compute this DimXpert position geometric tolerance.

.NET Syntax

Visual Basic (Declaration)

```
Function GetPlanarZoneVector( _  
    ByRef I As System.Double, _  
    ByRef J As System.Double, _  
    ByRef K As System.Double _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISwDMDimXpertPositionGeoTol  
Dim I As System.Double  
Dim J As System.Double  
Dim K As System.Double  
Dim value As System.Boolean  
  
value = instance.GetPlanarZoneVector(I, J, K)
```

C#

```
System.bool GetPlanarZoneVector(  
    out System.double I,  
    out System.double J,  
    out System.double K  
)
```

C++/CLI

```
System.bool GetPlanarZoneVector(  
& [Out] System.double I,  
& [Out] System.double J,  
& [Out] System.double K  
)
```

Parameters

I
i component of the planar zone vector
J
j component of the planar zone vector
K
k component of the planar zone vector

Return Value

True if method call is successful; false otherwise

Visual Basic for Applications (VBA) Syntax

See [SwDMDimXpertPositionGeoTol::GetPlanarZoneVector](#).

Example

See the examples on the interface page.

See Also

[ISwDMDimXpertPositionGeoTol Interface](#)
[ISwDMDimXpertPositionGeoTol Members](#)

Availability

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GetProjectedZone Method (ISwDMDimXpertPositionGeoTol)

[See Also](#)



[SolidWorks.Interop.swdocumentmgr Namespace](#) > [ISwDMDimXpertPositionGeoTol Interface](#) : GetProjectedZone Method (ISwDMDimXpertPositionGeoTol)

Gets the projected zone value of this DimXpert position geometric tolerance.

.NET Syntax

Visual Basic (Declaration)

```
Function GetProjectedZone( _  
    ByRef Enabled As System.Boolean, _  
    ByRef Value As System.Double _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISwDMDimXpertPositionGeoTol  
Dim Enabled As System.Boolean  
Dim Value As System.Double  
Dim value As System.Boolean  
  
value = instance.GetProjectedZone(Enabled, Value)
```

C#

```
System.bool GetProjectedZone(  
    out System.bool Enabled,  
    out System.double Value  
)
```

C++/CLI

```
System.bool GetProjectedZone(  
& [Out] System.bool Enabled,  
& [Out] System.double Value  
)
```

Parameters

Enabled

True if the projected zone is in effect; false otherwise

Value

Position projected zone value

Return Value

True if method call is successful; false otherwise

Visual Basic for Applications (VBA) Syntax

See [SwDMDimXpertPositionGeoTol::GetProjectedZone](#).

Example

See the examples on the interface page.

See Also

[ISwDMDimXpertPositionGeoTol Interface](#)
[ISwDMDimXpertPositionGeoTol Members](#)

Availability

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Modifier Property (ISwDMDimXpertPositionGeoTol)

[See Also](#)



[SolidWorks.Interop.swdocumentmgr Namespace](#) > [ISwDMDimXpertPositionGeoTol Interface](#) : Modifier Property (ISwDMDimXpertPositionGeoTol)

Gets the material condition modifier for this DimXpert position geometric tolerance.

.NET Syntax

Visual Basic (Declaration)

ReadOnly Property Modifier As [swDmDimXpertMaterialConditionModifier_e](#)

Visual Basic (Usage)



```
Dim instance As ISwDMDimXpertPositionGeoTol
Dim value As swDmDimXpertMaterialConditionModifier\_e

value = instance.Modifier
```

C#

```
swDmDimXpertMaterialConditionModifier\_e Modifier {get;}
```

C++/CLI

```
property swDmDimXpertMaterialConditionModifier\_e Modifier {
    swDmDimXpertMaterialConditionModifier\_e get();
}
```

Property Value

Material condition modifier as defined in [swDmDimXpertMaterialConditionModifier_e](#)

Visual Basic for Applications (VBA) Syntax

See [SwDMDimXpertPositionGeoTol::Modifier](#).

Example

See the examples on the interface page.

See Also

[ISwDMDimXpertPositionGeoTol Interface](#)
[ISwDMDimXpertPositionGeoTol Members](#)

Availability

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ZoneType Property (ISwDMDimXpertPositionGeoTol)

[See Also](#)



[SolidWorks.Interop.swdocumentmgr Namespace](#) > [ISwDMDimXpertPositionGeoTol Interface](#) : ZoneType Property (ISwDMDimXpertPositionGeoTol)

Gets the zone type of this DimXpert position geometric tolerance.

.NET Syntax

Visual Basic (Declaration)

ReadOnly Property ZoneType As [swDmDimXpertPositionZoneType_e](#)

Visual Basic (Usage)



```
Dim instance As ISwDMDimXpertPositionGeoTol
Dim value As swDmDimXpertPositionZoneType\_e

value = instance.ZoneType
```

C#

```
swDmDimXpertPositionZoneType\_e ZoneType {get;}
```

C++/CLI

```
property swDmDimXpertPositionZoneType\_e ZoneType {  
    swDmDimXpertPositionZoneType\_e get();  
}
```

Property Value

Zone type for the position geometric tolerance as defined in [swDmDimXpertPositionZoneType_e](#)

Visual Basic for Applications (VBA) Syntax

See [SwDMDimXpertPositionGeoTol::ZoneType](#).

Example

See the examples on the interface page.

See Also

[ISwDMDimXpertPositionGeoTol Interface](#)
[ISwDMDimXpertPositionGeoTol Members](#)

Availability

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ISwDMDimXpertRadiusDimTol (Type)

ISwDMDimXpertRadiusDimTol Interface

[See Also](#) [Members](#) [Example](#)



[SolidWorks.Interop.swdocumentmgr.Namespace](#) : ISwDMDimXpertRadiusDimTol Interface

Allows you to access tolerance information for DimXpert radius dimensions.

NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

.NET Syntax

Visual Basic (Declaration)

```
Public Interface ISwDMDimXpertRadiusDimTol
```

Visual Basic (Usage)



```
Dim instance As ISwDMDimXpertRadiusDimTol
```

C#

```
public interface ISwDMDimXpertRadiusDimTol
```

C++/CLI

```
public interface class ISwDMDimXpertRadiusDimTol
```

Visual Basic for Applications (VBA) Syntax

See [SwDMDimXpertRadiusDimTol](#).

Example

[Get DimXpert Tolerance2 \(VB.NET\)](#)
[Get DimXpert Tolerance2 \(C#\)](#)

Remarks

You can access more general information from [ISwDMDimXpertDimensionTolerance](#).

Accessors

[ISwDMDimXpertPart::GetAnnotations](#) and [ISwDMDimXpertPart::!GetAnnotations](#)

See Also

[ISwDMDimXpertRadiusDimTol Members](#)
[SolidWorks.Interop.swdocumentmgr.Namespace](#)

ISwDMDimXpertRadiusDimTol Interface Members

[See Also](#) [Properties](#)



[SolidWorks.Interop.swdocumentmgr.Namespace](#) : ISwDMDimXpertRadiusDimTol Interface

The following tables list the members exposed by [ISwDMDimXpertRadiusDimTol](#).

Public Properties

Name	Description
PatternTreatment	Gets the pattern treatment for this DimXpert radius dimension tolerance.

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See Also

[ISwDMDimXpertRadiusDimTol Interface](#)
[SolidWorks.Interop.swdocumentmgr Namespace](#)

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ISwDMDimXpertRadiusDimTol Interface Properties

[See Also](#)

[SolidWorks.Interop.swdocumentmgr Namespace](#) : ISwDMDimXpertRadiusDimTol Interface

For a list of all members of this type, see [ISwDMDimXpertRadiusDimTol members](#).

Public Properties

Name	Description
 PatternTreatment	Gets the pattern treatment for this DimXpert radius dimension tolerance.

[Top](#)

See Also

[ISwDMDimXpertRadiusDimTol Interface](#)
[SolidWorks.Interop.swdocumentmgr Namespace](#)

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PatternTreatment Property (ISwDMDimXpertRadiusDimTol)

[See Also](#)



[SolidWorks.Interop.swdocumentmgr Namespace](#) > [ISwDMDimXpertRadiusDimTol Interface](#) : PatternTreatment Property (ISwDMDimXpertRadiusDimTol)

Gets the pattern treatment for this DimXpert radius dimension tolerance.

.NET Syntax

Visual Basic (Declaration)

ReadOnly Property PatternTreatment As [swDmDimXpertPatternTreatmentType_e](#)

Visual Basic (Usage)



```
Dim instance As ISwDMDimXpertRadiusDimTol  
Dim value As swDmDimXpertPatternTreatmentType\_e  
  
value = instance.PatternTreatment
```

C#

```
swDmDimXpertPatternTreatmentType\_e PatternTreatment {get;}
```

C++/CLI

```
property swDmDimXpertPatternTreatmentType\_e PatternTreatment {  
    swDmDimXpertPatternTreatmentType\_e get();  
}
```

Property Value

Pattern treatment for the radius dimension tolerance as defined in [swDmDimXpertPatternTreatmentType_e](#)

Visual Basic for Applications (VBA) Syntax

See [SwDMDimXpertRadiusDimTol::PatternTreatment](#).

Example

See the examples on the interface page.

See Also

Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

ISwDMDimXpertSphereFeature (Type)

ISwDMDimXpertSphereFeature Interface

[See Also](#) [Members](#) [Example](#)



[SolidWorks.Interop.swdocumentmgr Namespace](#) : ISwDMDimXpertSphereFeature Interface

Allows you to access specific information about DimXpert spheres.

NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

.NET Syntax

Visual Basic (Declaration)

Public Interface ISwDMDimXpertSphereFeature

Visual Basic (Usage)



Dim instance As [ISwDMDimXpertSphereFeature](#)

C#

public interface ISwDMDimXpertSphereFeature

C++/CLI

public interface class ISwDMDimXpertSphereFeature

Visual Basic for Applications (VBA) Syntax

See [SwDMDimXpertSphereFeature](#).

Example

[Get DimXpert Sphere and Datum \(VB.NET\)](#)
[Get DimXpert Sphere and Datum \(C#\)](#)

Remarks

You can access more general information from [ISwDMDimXpertFeature](#).

Accessors

[ISwDMDimXpertPart::GetFeatures](#) and [ISwDMDimXpertPart::IGetFeatures](#)

See Also

[ISwDMDimXpertSphereFeature Members](#)
[SolidWorks.Interop.swdocumentmgr Namespace](#)

ISwDMDimXpertSphereFeature Interface Members

[See Also](#) [Properties](#) [Methods](#)



[SolidWorks.Interop.swdocumentmgr Namespace](#) : ISwDMDimXpertSphereFeature Interface

The following tables list the members exposed by [ISwDMDimXpertSphereFeature](#).

Public Properties

Name	Description
Inner	Gets whether this DimXpert sphere is a hole or a pin.

[Top](#)

Public Methods

Name	Description
 GetNominalSphere	Gets the origin coordinates and radius of this DimXpert sphere.

[Top](#)

See Also

[ISwDMDimXpertSphereFeature Interface](#)
[SolidWorks.Interop.swdocumentmgr Namespace](#)

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ISwDMDimXpertSphereFeature Interface Methods

[See Also](#)



[SolidWorks.Interop.swdocumentmgr Namespace](#) : ISwDMDimXpertSphereFeature Interface

For a list of all members of this type, see [ISwDMDimXpertSphereFeature members](#).

Public Methods

Name	Description
 GetNominalSphere	Gets the origin coordinates and radius of this DimXpert sphere.

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See Also

[ISwDMDimXpertSphereFeature Interface](#)
[SolidWorks.Interop.swdocumentmgr Namespace](#)

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ISwDMDimXpertSphereFeature Interface Properties

[See Also](#)



[SolidWorks.Interop.swdocumentmgr Namespace](#) : ISwDMDimXpertSphereFeature Interface

For a list of all members of this type, see [ISwDMDimXpertSphereFeature members](#).

Public Properties

Name	Description
 Inner	Gets whether this DimXpert sphere is a hole or a pin.

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See Also

[ISwDMDimXpertSphereFeature Interface](#)
[SolidWorks.Interop.swdocumentmgr Namespace](#)

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GetNominalSphere Method (ISwDMDimXpertSphereFeature)

[See Also](#)



[SolidWorks.Interop.swdocumentmgr Namespace](#) > [ISwDMDimXpertSphereFeature Interface](#) : GetNominalSphere Method (ISwDMDimXpertSphereFeature)

Gets the origin coordinates and radius of this DimXpert sphere.

.NET Syntax

Visual Basic (Declaration)

```
Function GetNominalSphere( _  
    ByRef R As System.Double, _  
    ByRef X As System.Double, _  
    ByRef Y As System.Double, _  
    ByRef Z As System.Double _  
) As System.Boolean
```

```

Dim instance As ISwDMDimXpertSphereFeature
Dim R As System.Double
Dim X As System.Double
Dim Y As System.Double
Dim Z As System.Double
Dim value As System.Boolean

value = instance.GetNominalSphere(R, X, Y, Z)

```

C#

```

System.bool GetNominalSphere(
    out System.double R,
    out System.double X,
    out System.double Y,
    out System.double Z
)

```

C++/CLI

```

System.bool GetNominalSphere(
& [Out] System.double R,
& [Out] System.double X,
& [Out] System.double Y,
& [Out] System.double Z
)

```

Parameters

R
Radius of the sphere

X
x coordinate of the sphere origin

Y
y coordinate of the sphere origin

Z
z coordinate of the sphere origin

Return Value

True if method call is successful; false otherwise

 Visual Basic for Applications (VBA) Syntax

See [SwDMDimXpertSphereFeature::GetNominalSphere](#).

 Example

See the examples on the interface page.

 See Also

[ISwDMDimXpertSphereFeature Interface](#)
[ISwDMDimXpertSphereFeature Members](#)

 Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

Inner Property (ISwDMDimXpertSphereFeature)

[See Also](#)



[SolidWorks.Interop.swdocumentmgr Namespace](#) > [ISwDMDimXpertSphereFeature Interface](#) : Inner Property (ISwDMDimXpertSphereFeature)

Gets whether this DimXpert sphere is a hole or a pin.

 .NET Syntax

ReadOnly Property Inner As System.Boolean

```
Dim instance As ISwDMDimXpertSphereFeature  
Dim value As System.Boolean  
  
value = instance.Inner
```

```
System.bool Inner {get;}
```

```
property System.bool Inner {  
    System.bool get();  
}
```

Property Value

True if the sphere is a hole; false if it's a pin

Visual Basic for Applications (VBA) Syntax

See [SwDMDimXpertSphereFeature::Inner](#).

Example

See the examples on the interface page.

See Also

[ISwDMDimXpertSphereFeature Interface](#)
[ISwDMDimXpertSphereFeature Members](#)

Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

ISwDMDimXpertStraightnessGeoTol (Type)

ISwDMDimXpertStraightnessGeoTol Interface

[See Also](#) [Members](#) [Example](#)



[SolidWorks.Interop.swdocumentmgr.Namespace](#) : ISwDMDimXpertStraightnessGeoTol Interface

Allows you to access DimXpert straightness geometric tolerance information.

NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

.NET Syntax

Visual Basic (Declaration)

```
Public Interface ISwDMDimXpertStraightnessGeoTol
```

Visual Basic (Usage)

```
Dim instance As ISwDMDimXpertStraightnessGeoTol
```

C#

```
public interface ISwDMDimXpertStraightnessGeoTol
```

C++/CLI

```
public interface class ISwDMDimXpertStraightnessGeoTol
```

Visual Basic for Applications (VBA) Syntax

See [SwDMDimXpertStraightnessGeoTol](#).

Example

[Get DimXpert Tolerance4 \(VB.NET\)](#)
[Get DimXpert Tolerance4 \(C#\)](#)

Remarks

You can access more general information from [ISwDMDimXpertGeometricTolerance](#).

Accessors

[ISwDMDimXpertPart::GetAnnotations](#) and [ISwDMDimXpertPart::!GetAnnotations](#)

See Also

[ISwDMDimXpertStraightnessGeoTol Members](#)
[SolidWorks.Interop.swdocumentmgr.Namespace](#)

ISwDMDimXpertStraightnessGeoTol Interface Members

[See Also](#) [Properties](#) [Methods](#)



[SolidWorks.Interop.swdocumentmgr.Namespace](#) : ISwDMDimXpertStraightnessGeoTol Interface

The following tables list the members exposed by [ISwDMDimXpertStraightnessGeoTol](#).

Public Properties

Name	Description
Modifier	Gets the material condition modifier for this DimXpert straightness geometric tolerance.
ZoneType	Gets the zone type of this DimXpert straightness geometric tolerance.

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Public Methods

Name	Description
 GetMaxTolerance	Gets the maximum tolerance for this DimXpert straightness geometric tolerance.
 GetPerUnitLength	Gets the per unit length for this DimXpert straightness geometric tolerance.
 GetPlanarZoneVector	Gets the planar zone vector that is used to compute this DimXpert straightness geometric tolerance.

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See Also

[ISwDMDimXpertStraightnessGeoTol Interface](#)
[SolidWorks.Interop.swdocumentmgr Namespace](#)

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ISwDMDimXpertStraightnessGeoTol Interface Methods

[See Also](#)



[SolidWorks.Interop.swdocumentmgr Namespace](#) : ISwDMDimXpertStraightnessGeoTol Interface

For a list of all members of this type, see [ISwDMDimXpertStraightnessGeoTol members](#).

Public Methods

Name	Description
 GetMaxTolerance	Gets the maximum tolerance for this DimXpert straightness geometric tolerance.
 GetPerUnitLength	Gets the per unit length for this DimXpert straightness geometric tolerance.
 GetPlanarZoneVector	Gets the planar zone vector that is used to compute this DimXpert straightness geometric tolerance.

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See Also

[ISwDMDimXpertStraightnessGeoTol Interface](#)
[SolidWorks.Interop.swdocumentmgr Namespace](#)

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ISwDMDimXpertStraightnessGeoTol Interface Properties

[See Also](#)



[SolidWorks.Interop.swdocumentmgr Namespace](#) : ISwDMDimXpertStraightnessGeoTol Interface

For a list of all members of this type, see [ISwDMDimXpertStraightnessGeoTol members](#).

Public Properties

Name	Description
 Modifier	Gets the material condition modifier for this DimXpert straightness geometric tolerance.
 ZoneType	Gets the zone type of this DimXpert straightness geometric tolerance.

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See Also

[ISwDMDimXpertStraightnessGeoTol Interface](#)
[SolidWorks.Interop.swdocumentmgr Namespace](#)

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GetMaxTolerance Method (ISwDMDimXpertStraightnessGeoTol)

[See Also](#)



[SolidWorks.Interop.swdocumentmgr Namespace](#) > [ISwDMDimXpertStraightnessGeoTol Interface](#) : GetMaxTolerance Method (ISwDMDimXpertStraightnessGeoTol)

Gets the maximum tolerance for this DimXpert straightness geometric tolerance.

.NET Syntax

Visual Basic (Declaration)

```
Function GetMaxTolerance( _  
    ByRef IsMax As System.Boolean, _  
    ByRef Tolerance As System.Double _  
    ) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISwDMDimXpertStraightnessGeoTol  
Dim IsMax As System.Boolean  
Dim Tolerance As System.Double  
Dim value As System.Boolean  
  
value = instance.GetMaxTolerance(IsMax, Tolerance)
```

C#

```
System.bool GetMaxTolerance(  
    out System.bool IsMax,  
    out System.double Tolerance  
)
```

C++/CLI

```
System.bool GetMaxTolerance(  
& [Out] System.bool IsMax,  
& [Out] System.double Tolerance  
)
```

Parameters

IsMax
True, if maximum tolerance is set; false, otherwise

Tolerance
Maximum tolerance

Return Value

True if method call is successful; false otherwise

Visual Basic for Applications (VBA) Syntax

See [SwDMDimXpertStraightnessGeoTol::GetMaxTolerance](#).

Example

See the examples on the interface page.

See Also

[ISwDMDimXpertStraightnessGeoTol Interface](#)
[ISwDMDimXpertStraightnessGeoTol Members](#)

Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

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GetPerUnitLength Method (ISwDMDimXpertStraightnessGeoTol)

[See Also](#)



[SolidWorks.Interop.swdocumentmgr Namespace](#) > [ISwDMDimXpertStraightnessGeoTol Interface](#) : GetPerUnitLength Method (ISwDMDimXpertStraightnessGeoTol)

Gets the per unit length for this DimXpert straightness geometric tolerance.

.NET Syntax

Visual Basic (Declaration)

```
Function GetPerUnitLength( _
    ByRef Enabled As System.Boolean, _
    ByRef Distance As System.Double _
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISwDMDimXpertStraightnessGeoTol
Dim Enabled As System.Boolean
Dim Distance As System.Double
Dim value As System.Boolean

value = instance.GetPerUnitLength(Enabled, Distance)
```

C#

```
System.bool GetPerUnitLength(
    out System.bool Enabled,
    out System.double Distance
)
```

C++/CLI

```
System.bool GetPerUnitLength(
    & [Out] System.bool Enabled,
    & [Out] System.double Distance
)
```

Parameters

Enabled

True if per unit length straightness tolerance is in effect; false if not

Distance

Length for per unit length

Return Value

True if method call is successful; false otherwise

Visual Basic for Applications (VBA) Syntax

See [SwDMDimXpertStraightnessGeoTol::GetPerUnitLength](#).

Example

See the examples on the interface page.

See Also

[ISwDMDimXpertStraightnessGeoTol Interface](#)
[ISwDMDimXpertStraightnessGeoTol Members](#)

Availability

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GetPlanarZoneVector Method (ISwDMDimXpertStraightnessGeoTol)

[See Also](#)



[SolidWorks.Interop.swdocumentmgr Namespace](#) > [ISwDMDimXpertStraightnessGeoTol Interface](#) : GetPlanarZoneVector Method (ISwDMDimXpertStraightnessGeoTol)

Gets the planar zone vector that is used to compute this DimXpert straightness geometric tolerance.

.NET Syntax

Visual Basic (Declaration)

```
Function GetPlanarZoneVector( _
    ByRef I As System.Double, _
    ByRef J As System.Double, _
    ByRef K As System.Double _
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISwDMDimXpertStraightnessGeoTol
Dim I As System.Double
Dim J As System.Double
Dim K As System.Double
Dim value As System.Boolean

value = instance.GetPlanarZoneVector(I, J, K)
```

C#

```
System.bool GetPlanarZoneVector(
    out System.double I,
    out System.double J,
    out System.double K
)
```

C++/CLI

```
System.bool GetPlanarZoneVector(
    & [Out] System.double I,
    & [Out] System.double J,
    & [Out] System.double K
)
```

Parameters

I
i component of planar zone vector
J
j component of planar zone vector
K
k component of planar zone vector

Return Value

True if method call is successful; false otherwise

Visual Basic for Applications (VBA) Syntax

See [SwDMDimXpertStraightnessGeoTol::GetPlanarZoneVector](#).

Example

See the examples on the interface page.

See Also

[ISwDMDimXpertStraightnessGeoTol Interface](#)
[ISwDMDimXpertStraightnessGeoTol Members](#)

Availability

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Modifier Property (ISwDMDimXpertStraightnessGeoTol)

[See Also](#)



[SolidWorks.Interop.swdocumentmgr Namespace](#) > [ISwDMDimXpertStraightnessGeoTol Interface](#) : Modifier Property (ISwDMDimXpertStraightnessGeoTol)

Gets the material condition modifier for this DimXpert straightness geometric tolerance.

Visual Basic Syntax

Visual Basic (Declaration)

ReadOnly Property Modifier As [swDmDimXpertMaterialConditionModifier_e](#)

Visual Basic (Usage)



```
Dim instance As ISwDMDimXpertStraightnessGeoTol  
Dim value As swDmDimXpertMaterialConditionModifier\_e  
  
value = instance.Modifier
```

C#

```
swDmDimXpertMaterialConditionModifier\_e Modifier {get;}
```

C++/CLI

```
property swDmDimXpertMaterialConditionModifier\_e Modifier {  
    swDmDimXpertMaterialConditionModifier\_e get();  
}
```

Property Value

Material condition modifier as defined in [swDmDimXpertMaterialConditionModifier_e](#)

Visual Basic for Applications (VBA) Syntax

See [SwDMDimXpertStraightnessGeoTol::Modifier](#).

Example

See the examples on the interface page.

See Also

[ISwDMDimXpertStraightnessGeoTol Interface](#)
[ISwDMDimXpertStraightnessGeoTol Members](#)

Availability

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ZoneType Property (ISwDMDimXpertStraightnessGeoTol)

[See Also](#)



[SolidWorks.Interop.swdocumentmgr Namespace](#) > [ISwDMDimXpertStraightnessGeoTol Interface](#) : ZoneType Property (ISwDMDimXpertStraightnessGeoTol)

Gets the zone type of this DimXpert straightness geometric tolerance.

Visual Basic Syntax

Visual Basic (Declaration)

ReadOnly Property ZoneType As [swDmDimXpertStraightnessZoneType_e](#)

Visual Basic (Usage)



```
Dim instance As ISwDMDimXpertStraightnessGeoTol  
Dim value As swDmDimXpertStraightnessZoneType\_e  
  
value = instance.ZoneType
```

C#

```
swDmDimXpertStraightnessZoneType\_e ZoneType {get;}
```

C++/CLI

```
property swDmDimXpertStraightnessZoneType\_e ZoneType {  
    swDmDimXpertStraightnessZoneType\_e get();  
}
```

Property Value

Zone type for the straightness geometric tolerance as defined in [swDmDimXpertStraightnessZoneType_e](#)

Visual Basic for Applications (VBA) Syntax

See [SwDMDimXpertStraightnessGeoTol::ZoneType](#).

Example

See the examples on the interface page.

See Also

[ISwDMDimXpertStraightnessGeoTol Interface](#)
[ISwDMDimXpertStraightnessGeoTol Members](#)

Availability

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ISwDMDimXpertSurfaceFeature (Type)

ISwDMDimXpertSurfaceFeature Interface

[See Also](#) [Members](#) [Example](#)



[SolidWorks.Interop.swdocumentmgr Namespace](#) : ISwDMDimXpertSurfaceFeature Interface

Allows you to access specific information about DimXpert surfaces.

NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

.NET Syntax

Visual Basic (Declaration)

Public Interface ISwDMDimXpertSurfaceFeature

Visual Basic (Usage)



Dim instance As [ISwDMDimXpertSurfaceFeature](#)

C#

public interface ISwDMDimXpertSurfaceFeature

C++/CLI

public interface class ISwDMDimXpertSurfaceFeature

Visual Basic for Applications (VBA) Syntax

See [SwDMDimXpertSurfaceFeature](#).

Example

[Get DimXpert Surface Feature \(VB.NET\)](#)
[Get DimXpert Surface Feature \(C#\)](#)

Remarks

You can access more general information from [ISwDMDimXpertFeature](#).

Accessors

[ISwDMDimXpertPart::GetFeatures](#) and [ISwDMDimXpertPart::IGetFeatures](#)

See Also

[ISwDMDimXpertSurfaceFeature Members](#)
[SolidWorks.Interop.swdocumentmgr Namespace](#)

ISwDMDimXpertSurfaceFeature Interface Members

[See Also](#)



[SolidWorks.Interop.swdocumentmgr Namespace](#) : ISwDMDimXpertSurfaceFeature Interface

The following tables list the members exposed by [ISwDMDimXpertSurfaceFeature](#).

See Also

[ISwDMDimXpertSurfaceFeature Interface](#)
[SolidWorks.Interop.swdocumentmgr Namespace](#)

ISwDMDimXpertSurfaceProfileGeoTol (Type)

ISwDMDimXpertSurfaceProfileGeoTol Interface

[See Also](#) [Members](#) [Example](#)



[SolidWorks.Interop.swdocumentmgr Namespace](#) : ISwDMDimXpertSurfaceProfileGeoTol Interface

Allows you to access DimXpert surface profile geometric tolerance information.

NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

.NET Syntax

Visual Basic (Declaration)

Public Interface ISwDMDimXpertSurfaceProfileGeoTol

Visual Basic (Usage)



Dim instance As [ISwDMDimXpertSurfaceProfileGeoTol](#)

C#

public interface ISwDMDimXpertSurfaceProfileGeoTol

C++/CLI

public interface class ISwDMDimXpertSurfaceProfileGeoTol

Visual Basic for Applications (VBA) Syntax

See [SwDMDimXpertSurfaceProfileGeoTol](#).

Example

[Get DimXpert Tolerance1 \(VB.NET\)](#)
[Get DimXpert Tolerance1 \(C#\)](#)

Remarks

You can access more general information from [ISwDMDimXpertGeometricTolerance](#).

Accessors

[ISwDMDimXpertPart::GetAnnotations](#) and [ISwDMDimXpertPart::IGetAnnotations](#)

See Also

[ISwDMDimXpertSurfaceProfileGeoTol Members](#)
[SolidWorks.Interop.swdocumentmgr Namespace](#)

ISwDMDimXpertSurfaceProfileGeoTol Interface Members

[See Also](#) [Properties](#)



[SolidWorks.Interop.swdocumentmgr Namespace](#) : ISwDMDimXpertSurfaceProfileGeoTol Interface

The following tables list the members exposed by [ISwDMDimXpertSurfaceProfileGeoTol](#).

Public Properties

Name	Description
OuterToleranceValue	Gets the outer tolerance value for this DimXpert surface profile geometric tolerance.

[Top](#)

See Also

[ISwDMDimXpertSurfaceProfileGeoTol Interface](#)
[SolidWorks.Interop.swdocumentmgr Namespace](#)

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ISwDMDimXpertSurfaceProfileGeoTol Interface Properties

[See Also](#)

 [SolidWorks.Interop.swdocumentmgr Namespace](#) : ISwDMDimXpertSurfaceProfileGeoTol Interface

For a list of all members of this type, see [ISwDMDimXpertSurfaceProfileGeoTol members](#).

Public Properties

Name	Description
 OuterToleranceValue	Gets the outer tolerance value for this DimXpert surface profile geometric tolerance.

[Top](#)

See Also

[ISwDMDimXpertSurfaceProfileGeoTol Interface](#)
[SolidWorks.Interop.swdocumentmgr Namespace](#)

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OuterToleranceValue Property (ISwDMDimXpertSurfaceProfileGeoTol)

[See Also](#)



[SolidWorks.Interop.swdocumentmgr Namespace](#) > [ISwDMDimXpertSurfaceProfileGeoTol Interface](#) : OuterToleranceValue Property (ISwDMDimXpertSurfaceProfileGeoTol)

Gets the outer tolerance value for this DimXpert surface profile geometric tolerance.

.NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property OuterToleranceValue As System.Double
```

Visual Basic (Usage)



```
Dim instance As ISwDMDimXpertSurfaceProfileGeoTol  
Dim value As System.Double  
  
value = instance.OuterToleranceValue
```

C#

```
System.double OuterToleranceValue {get;}
```

C++/CLI

```
property System.double OuterToleranceValue {  
    System.double get();  
}
```

Property Value

Outer tolerance value

Visual Basic for Applications (VBA) Syntax

See [SwDMDimXpertSurfaceProfileGeoTol.OuterToleranceValue](#).

Example

See the examples on the interface page.

See Also

Availability

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